MEDIUM

ADVANCED MATH

Advanced Math

16

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

$$D = 5{,}640(1.9)^t$$

The equation above estimates the global data traffic D, in terabytes, for the year that is t years after 2010. What is the best interpretation of the number 5,640 in this context?

A. The estimated amount of increase of data traffic, in terabytes, each year

B. The estimated percent increase in the data traffic, in terabytes, each year

C. The estimated data traffic, in terabytes, for the year that is t years after 2010

D. The estimated data traffic, in terabytes, in 2010

Q2 ADVANCED MATH

$$N(d) = 115(0.90)^d$$

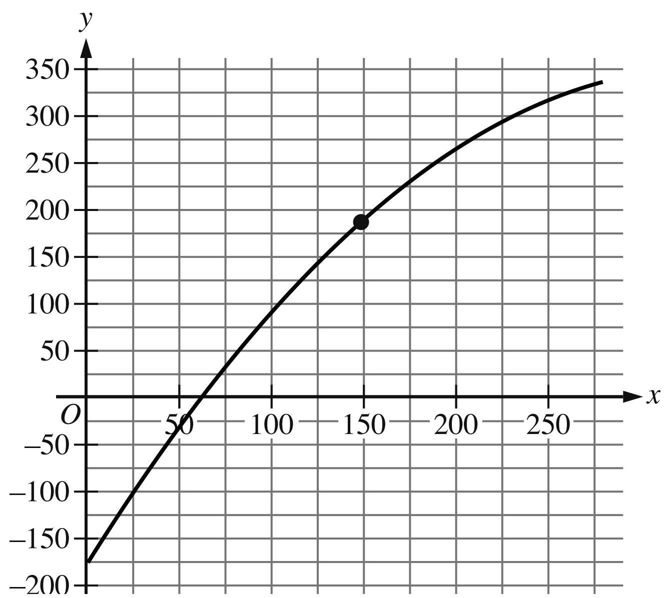
The function N defined above can be used to model the number of species of brachiopods at various ocean depths d, where d is in <u>hundreds</u> of meters. Which of the following does the model predict?

A. For every increase in depth by 1 meter, the number of brachiopod species decreases by 115.

B. For every increase in depth by 1 meter, the number of brachiopod species decreases by 10%.

C. For every increase in depth by 100 meters, the number of brachiopod species decreases by 115.

D. For every increase in depth by 100 meters, the number of brachiopod species decreases by 10%.



- The parabola opens downward.
- The parabola passes through the following approximate points:
  - (50 comma negative 33.83)
  - (149.02 comma 186.05)
  - (250 comma 315.37)

The graph shows the estimated boiling point y, in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $149.02186.05$?

A. A normal paraffin with a molecular weight of 186.05 grams per mole has an estimated boiling point of 149.02 degrees Celsius.

B. A normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.

C. The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

D. The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

Q4

ADVANCED MATH

A rectangle has a length of x units and a width of x - 15 units. If the rectangle has an area of 76 square units, what is the value of x?

A. 4

B. 19

C. 23

D. 76

Q5 ADVANCED MATH

$$ft = 5000.5^{\frac{t}{12}}$$

The function f models the intensity of an X-ray beam, in number of particles in the X-ray beam, t millimeters below the surface of a sample of iron. According to the model, what is the estimated number of particles in the X-ray beam when it is at the surface of the sample of iron?

A. 500

B. 12

C. 5

D. 2

$$f(x) = (x+4)(x-1)(2x-3)$$

The function f is defined above. Which of the following is NOT an x-intercept of the graph of the function in the xy-plane?

A. $(-4, 0)$

B. $\left(-\frac{2}{3}, 0\right)$

C. $(1, 0)$

D. $\left(\frac{3}{2}, 0\right)$

Q7 ADVANCED MATH

According to Moore’s law, the number of transistors included on microprocessors doubles every 2 years. In 1985, a microprocessor was introduced that had 275,000 transistors. Based on this information, in which of the following years does Moore’s law estimate the number of transistors to reach 1.1 million?

A. 1987

B. 1989

C. 1991

D. 1994

Q8 ADVANCED MATH

A company opens an account with an initial balance of $\$ 36{,}100.00$. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function A, where $A(t)$ is the account balance, in dollars, t years after the account is opened. The account balance after 13 years is $\$ 68{,}071.93$. Which equation could define A?

A. $At = 36{,}100.001.05^t$

B. $At = 31{,}971.931.05^t$

C. $At = 31{,}971.930.05^t$

D. $At = 36{,}100.000.05^t$

Q9 ADVANCED MATH

The function p is defined by $pn = 7n^3$. What is the value of n when $p(n)$ is equal to 56?

A. 2

B. $\frac{8}{3}$

C. 7

D. 8

Q10 ADVANCED MATH

$$gx = x^2 + 55$$

What is the minimum value of the given function?

A. 0

B. 55

C. 110

D. 3,025